

MVG-WAM: Multiple View Geometry-Aware World-Action Modeling for Robotic Manipulation

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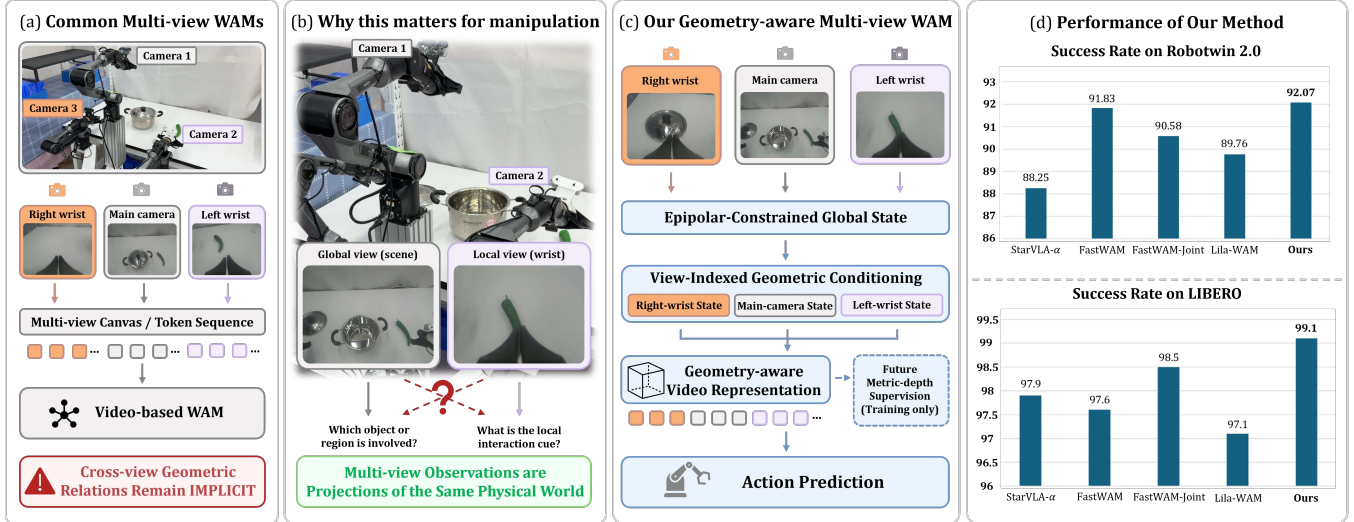


Fig. 1. **Motivation and conceptual overview of our MVG-WAM.** (a) Common multi-view interfaces tile images or concatenate tokens, leaving calibrated cross-view geometric relations implicit. (b) Scene and wrist views provide complementary global and local evidence about the same physical world. Manipulation therefore requires relating these observations to reason about object interaction. (c) MVG-WAM structures the representation of synchronized multi-view observations through an epipolar-constrained global state and view-indexed geometric conditioning, producing a geometry-aware video representation for action prediction. Future metric-depth supervision metrically grounds the geometry-aware video representation during training and is removed at deployment. (d) Among methods without additional embodied pretraining, MVG-WAM achieves strong average success rates on both LIBERO and RoboTwin 2.0.

Abstract—World-Action Models (WAMs) couple visual dynamics with action prediction, bringing the rich priors of pretrained video models to robotic manipulation. However, their multi-view interfaces typically tile images or concatenate tokens, leaving the geometric relationships among synchronized cameras implicit. This makes it harder to connect global scene context with the local geometry required for interaction. We introduce the **Multi-View Geometry-Aware World-Action Model (MVG-WAM)**, which organizes these observations as related projections of one physical world rather than separate images on a canvas. Our model combines an epipolar-constrained global state with view-indexed geometric states jointly inferred from synchronized observations. Camera-aware routing supplies each video region with its corresponding geometric context and the shared global state, explicitly structuring the representation used for action prediction. We further ground the geometry-aware representation in metric scale through multi-horizon future-depth supervision, without requiring depth decoding during action rollout. MVG-WAM achieves average success rates of 99.1% on LIBERO and 92.07% on RoboTwin 2.0, demonstrating competitive performance across both benchmarks. Real-world experiments on Cobot Magic further demonstrate a 91.3% success rate among 150 trials spanning three manipulation tasks.

I. INTRODUCTION

World-Action Models (WAMs) jointly model future visual dynamics and robot actions, allowing control to benefit from predictive representations of the environment [1], [2].

This paradigm has attracted growing attention for robotic manipulation. To exploit large-scale spatiotemporal priors, many recent WAMs build on pretrained video-generation backbones [3], [4], [5]. While this inheritance provides strong visual dynamics priors, it also carries over the representation interface of video generation.

This interface becomes problematic in multi-view manipulation. Robots commonly observe the workspace through a scene camera and multiple wrist cameras. The scene view provides global workspace context, while wrist views reveal local geometry around the gripper and contact regions. Crucially, these observations are synchronized projections of the same physical world. Existing WAMs often accommodate them by tiling multiple views on a single video canvas or concatenating their visual tokens [4], [6]. As illustrated in Fig. 1(a), these interfaces preserve each camera’s visual content but leave calibrated cross-view relations implicit. Manipulation nevertheless requires linking global scene context to local interaction geometry (Fig. 1(b)). The model must therefore recover cross-view correspondences from the training signal while simultaneously learning visual dynamics and actions. This additional burden can weaken geometric coherence across views and, in turn, spatial reasoning for action prediction.

To this end, we introduce the **Multi-View Geometry-**

Aware World–Action Model (MVG-WAM). Our design follows a simple principle: *multiple camera observations should be represented as geometrically related projections of one physical world, rather than merely as multiple images on a video canvas.* We therefore explicitly organize the multi-view representation around the physical relations among synchronized camera observations.

As outlined in Fig. 1(c), MVG-WAM realizes this principle through three complementary mechanisms. First, we construct a global state under explicit epipolar constraints, allowing information to be aggregated across geometrically related views. Second, a geometry foundation model jointly encodes the synchronized observations into correlated per-view states. We group the video tokens according to their camera regions. Each group is conditioned on the corresponding per-view state. This preserves view alignment while injecting multi-view geometric context. Together, the first two mechanisms provide relational multi-view geometry through calibrated cross-view aggregation and view-indexed conditioning. We complement this structure with multi-horizon metric-depth supervision, which anchors the geometry-aware video representation to physical scale. This metric grounding encourages the latent dynamics to preserve physically meaningful geometry as the scene evolves under actions.

Our contributions are fourfold:

- We formulate multi-view observations in WAMs as geometrically related projections of a shared physical world, and introduce an epipolar-constrained global state to model their shared structure.
- We introduce view-indexed geometric conditioning from jointly encoded multi-view features, linking each camera region of the video representation to its corresponding geometric state.
- We metrically ground the geometry-aware video representation through multi-horizon future-depth supervision, encouraging physically scaled geometry across views and time without requiring depth decoding at deployment.
- Without additional embodied pretraining, our MVG-WAM achieves competitive performance on both LIBERO and RoboTwin 2.0, together with strong real-world manipulation results.

II. RELATED WORK

A. World–Action Modeling

World–Action Models (WAMs) jointly model visual dynamics and robot actions. UVA [1] and UWM [2] established unified video–action learning through shared representations or diffusion formulations. Subsequent methods, including Motus [3], mimic-video [7], LingBot-VA [8], and DreamZero [4], further exploit large-scale video priors for robot control. FastWAM [6] shows that video co-training remains useful even without future-frame generation at deployment. Together, these works highlight video representations as an effective basis for action learning, while multi-view

observations are still commonly handled through generic video interfaces that leave calibrated cross-view relations implicit.

B. Geometry-Aware Robot Learning

Geometry has increasingly been introduced into robot learning to improve spatial reasoning and manipulation accuracy. For VLA policies, GeoAware-VLA [9] incorporates geometric visual features, while Spatial Forcing [10] and ROCKET [11] transfer spatial priors from pretrained 3D models. A similar trend appears in WAMs [12]: GeoSem-WAM [13] incorporates geometric and semantic cues, X-WAM [14] predicts multi-view RGB-D futures, and WAM4D [15] uses future depth as geometric supervision. MECo-WAM [16] and SG-WAM [17] further explore geometric priors for action-conditioned representations. These works mainly study *what geometric information* should be encoded or predicted; our focus is instead on *how synchronized camera observations should be geometrically related within a multi-view WAM.*

C. Multi-View Geometric Modeling

Multi-view geometry provides explicit mechanisms for relating different projections of a shared 3D scene. MVDiff [18], EpiDiff [19], and MVISTA-4D [20] use epipolar constraints or geometry-aware feature interaction to improve cross-view correspondence. In parallel, geometry foundation models such as VGGT [21], VGGT- Ω [22], and Depth Anything 3 [23] provide strong geometric priors from multi-view observations. These directions are complementary: calibrated geometry constrains where cross-view information is exchanged, while learned geometric representations capture scene structure across views. MVG-WAM brings these ideas into world–action modeling, organizing synchronized observations as geometrically related projections rather than independent regions of a video interface.

III. METHODOLOGY

A. WAM Formulation and Method Overview

We consider a video-based World–Action Model with a video expert and an action expert. At time t , the robot receives a language instruction c , a proprioceptive state s_t , and synchronized observations from V calibrated cameras:

$$\mathcal{O}_t = \{I_t^v, K_t^v, T_t^v\}_{v=1}^V. \quad (1)$$

Here, I_t^v is the RGB observation and K_t^v is the intrinsic matrix. $T_t^v \in SE(3)$ maps world-frame points to camera v . The synchronized images are arranged into a multi-view canvas, denoted by X_t .

During training, the model predicts a future canvas sequence $X_{t+1:t+H_v}$ and an action trajectory $\mathbf{a}_{t:t+H_a-1}$. A video VAE maps the canvas sequence into latent video tokens. The current canvas remains a clean first-frame condition. Future video latents and action tokens are independently perturbed at diffusion timesteps τ_v and τ_a . The two experts then predict their corresponding flow-matching targets.

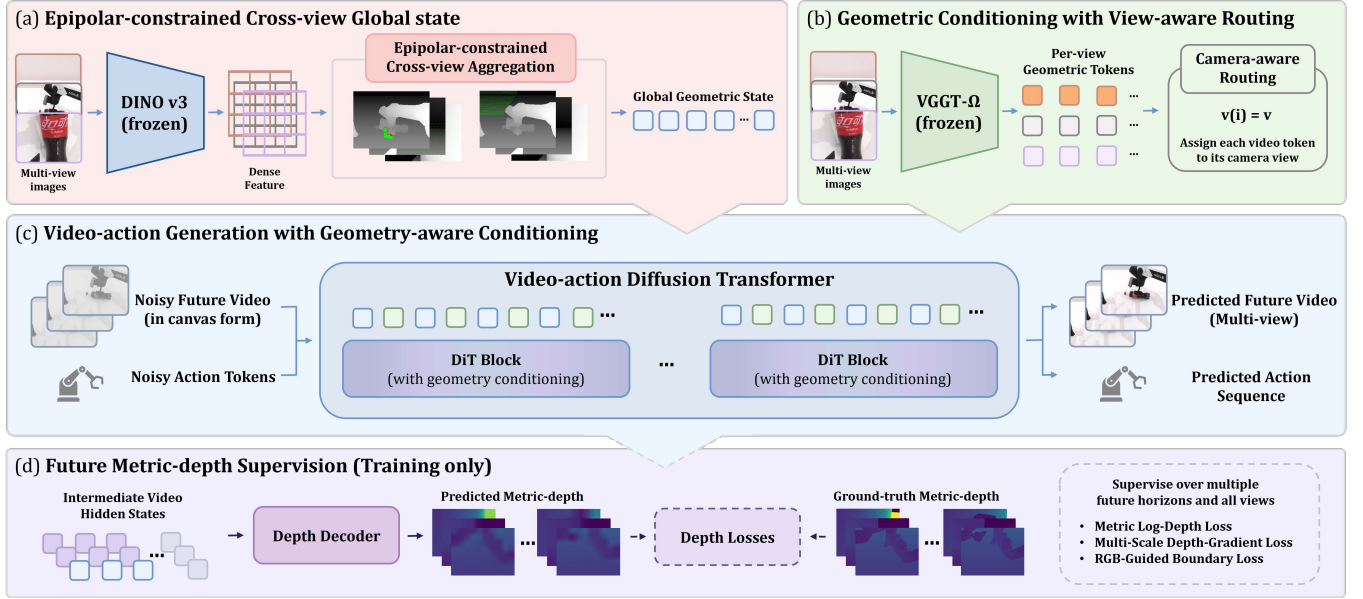


Fig. 2. **Architecture of our MVG-WAM.** (a) Dense DINOv3 features are aggregated under calibrated epipolar constraints to provide cross-view geometric context. (b) Compact VGGT- Ω registers form view-indexed geometric states, which are routed to the corresponding camera regions. (c) The global and view-indexed states condition the video stream of the video-action diffusion transformer, while action tokens access the geometry-enhanced video representation through mixed attention. (d) During training, future metric-depth supervision provides metric-scale grounding for intermediate denoising features across views and horizons; the depth decoder is not used during action rollout.

Action tokens attend to the full video sequence through mixed attention. Language and proprioception provide additional conditioning. The base training objective contains a video loss $\mathcal{L}_{\text{vid}}$ and an action loss $\mathcal{L}_{\text{act}}$.

Figure 2 summarizes the architecture of MVG-WAM. As shown in Fig. 2(a), calibrated epipolar geometry constrains cross-view feature aggregation into a compact global state. Figure 2(b) introduces view-indexed geometric states and camera-aware routing for the corresponding canvas regions. These two geometric conditions are integrated into the video-action backbone in Fig. 2(c). Finally, Fig. 2(d) provides metric-scale grounding through future-depth supervision during training.

MVG-WAM retains the video interface while injecting calibrated cross-view and view-indexed geometric structure into its latent representation.

B. Epipolar-Constrained Cross-View Global State

The video backbone operates on compact VAE latents optimized for efficient generation. Such strongly compressed representations can limit information capacity and fine-grained visual structure [24]. We therefore perform geometry-constrained cross-view retrieval in a dense feature space.

For each current observation, we extract a dense DINOv3 patch feature map [25]. We denote the projected feature map by $S_i^v \in \mathbb{R}^{H_f \times W_f \times d_s}$. All cameras share the same feature projector. A learned view embedding identifies the source camera of each feature map.

The camera calibration must remain consistent with image preprocessing. Let A_v map coordinates from the original image to the dense encoder input. For readability, we omit

the time index from camera quantities within this subsection. The feature-space geometry is

$$\begin{aligned} \bar{K}_v &= A_v K_v, \\ T_{sq} &= T_s T_q^{-1} = [R_{sq} \mid t_{sq}], \\ E_{sq} &= [t_{sq}] \times R_{sq}, \\ F_{qs} &= \bar{K}_s^{-\top} E_{sq} \bar{K}_q^{-1}. \end{aligned} \quad (2)$$

For a query patch center $\tilde{p}$, the corresponding source observation must lie on $\ell_{qs}(p) = F_{qs} \tilde{p}$. The epipolar line restricts correspondence search to a geometrically admissible region, within which attention selects candidate features.

We clip each epipolar line to the valid source feature region. A narrow band is sampled along the resulting in-image segment. The continuous locations are converted to feature-grid coordinates, and source features are obtained by bilinear interpolation.

A query is marked valid, $m_{qs}(p) = 1$, when the epipolar line intersects the source feature region and contains at least one valid sampled candidate.

Following epipolar-constrained multi-view interaction [19], the query feature attends to its sampled candidates using masked multi-head attention. The attention includes a learned candidate-position bias. A learnable null candidate provides a no-match option. The resulting attended feature is denoted by $U_{qs}(p)$.

We compute a confidence value $c_{qs}(p) = \sigma(g([S_i^q(p); U_{qs}(p)]))$. The directed cross-view residual is

$$\Delta S_{qs}(p) = m_{qs}(p) c_{qs}(p) U_{qs}(p). \quad (3)$$

Queries without valid real candidates receive an exact zero

update. The null candidate and confidence gate reduce the effect of unsupported matches.

We apply the same operator to all valid ordered camera pairs. Each view can therefore act as both query and source. Let $\mathcal{V}_q(p) = \{s \neq q \mid m_{qs}(p) = 1\}$ be the valid source-view set. The enhanced query feature is

$$\tilde{S}_t^q(p) = S_t^q(p) + \frac{1}{|\mathcal{V}_q(p)|} \sum_{s \in \mathcal{V}_q(p)} \Delta S_{qs}(p). \quad (4)$$

The residual term is set to zero when $\mathcal{V}_q(p)$ is empty.

We concatenate the enhanced features into $M_t^{\text{epi}} = [\tilde{S}_t^1; \dots; \tilde{S}_t^V]$. A learnable query bank pools this memory into a compact residual:

$$\begin{aligned} \Delta G_t^{\text{epi}} &= \text{CAStack}(Q_{\text{epi}}, M_t^{\text{epi}}), \\ G_t &= G_t^{\text{base}} + \alpha_{\text{epi}} \Delta G_t^{\text{epi}}. \end{aligned} \quad (5)$$

CAStack denotes a small stack of multi-head cross-attention and feed-forward blocks. G_t^{base} is the compact geometry memory introduced in Sec. III-C. The residual gate α_{epi} is learned during training.

Query pooling compresses the epipolar-enhanced dense features into a compact cross-view memory ΔG_t^{epi} .

C. View-Indexed Geometric Conditioning

The epipolar branch provides dense cross-view evidence. We additionally require compact camera-level context for each input view. To this end, a frozen geometry foundation model jointly processes the synchronized observations. We instantiate this encoder with VGGT- Ω [22].

For every input view, VGGT- Ω produces a camera-and-register token sequence. We discard the camera token and retain the compact view-indexed register bank R_t^v .

A shared adapter maps the registers into the WAM feature space. A learnable query bank then summarizes all adapted states:

$$\begin{aligned} P_t^v &= \mathcal{A}_{\text{reg}}(R_t^v) + e_v, \\ P_t &= [P_t^1; \dots; P_t^V], \\ G_t^{\text{base}} &= \text{CAStack}(Q_{\text{geo}}, P_t). \end{aligned} \quad (6)$$

The embedding e_v identifies the camera stream, while current camera geometry enters through the calibrated epipolar branch.

Q_{geo} and Q_{epi} use matched slot counts and dimensions, allowing the epipolar branch to act as a residual correction.

We next associate latent video tokens with their camera regions. Let $v(i)$ denote the region containing the latent-grid center of token i . This spatial association is repeated across all latent temporal groups. The current state P_t^v therefore conditions all temporal tokens associated with region v .

The routing assigns each latent token to the camera region containing its latent-grid center.

Let $\tilde{h}_i^{(l)}$ denote a video token after mixed attention at layer l . We define the view and global conditions by

$$\begin{aligned} C_{i,v}^{(l)} &= \text{CA}_v^{(l)}(\tilde{h}_i^{(l)}, P_t^{v(i)}), \\ C_{i,g}^{(l)} &= \text{CA}_g^{(l)}(\tilde{h}_i^{(l)}, G_t). \end{aligned} \quad (7)$$

The conditioned token is

$$\hat{h}_i^{(l)} = \tilde{h}_i^{(l)} + \alpha_v^{(l)} C_{i,v}^{(l)} + \alpha_g^{(l)} C_{i,g}^{(l)}. \quad (8)$$

The two branches use independent cross-attention parameters. Their layer-specific residual gates are initialized to zero.

Geometric conditioning is inserted after mixed attention and before text cross-attention and the block FFN throughout the video expert. Action tokens subsequently access the geometry-enhanced video representation through mixed-attention layers.

D. Future Metric-Depth Supervision

Epipolar aggregation and view-indexed conditioning establish relational geometry across synchronized observations. To anchor this geometry-aware representation to physical scale, we supervise intermediate denoising features with metric depth at multiple future horizons. This metric grounding complements the cross-view structure by encouraging the evolving video representation to preserve absolute spatial scale.

Let $\Delta = \{\delta_1, \dots, \delta_{H_d}\}$ denote the supervised future horizons. For camera v and horizon δ_h , the target $D_{t+\delta_h}^{v,*}$ is camera-axis depth in the corresponding future camera frame. All targets are expressed in the same physical unit, providing a shared metric reference for the view-indexed geometry across future horizons.

Metric depth is used as auxiliary training supervision, while rollout relies only on RGB, proprioception, language, and calibration.

We collect complete post-block video states from a set of layers $\mathcal{L}$. These states have passed through mixed attention, MVG conditioning, text cross-attention, and the block FFN. The latent canvas is divided using the same camera-region layout as the conditioning module.

Each selected feature is normalized and projected to a common decoder dimension. Multi-level features from the same camera region are fused by a shared convolutional decoder. View and temporal-group embeddings distinguish the outputs. The video diffusion timestep is injected through FiLM modulation [31].

Let $H_{\tau_v}^v = \{H_{\tau_v}^{(l),v}\}_{l \in \mathcal{L}}$ collect the selected features for view v . The decoder produces log-depth and metric depth as

$$\begin{aligned} \hat{\mathbf{d}}^v &= \mathcal{D}(H_{\tau_v}^v, \tau_v, v), \\ \hat{\mathbf{D}}^v &= \exp(\text{clip}(\hat{\mathbf{d}}^v)). \end{aligned} \quad (9)$$

The entries of $\hat{\mathbf{D}}^v$ follow the horizon order in Δ . The clipping interval $[d_{\min}, d_{\max}]$ is defined in log-depth space. Invalid pixels and unavailable horizons are excluded by validity masks.

Because the decoder reads denoising features produced from noised future latents, the depth objective acts as metric grounding for the evolving representation rather than as a standalone forecasting module.

The auxiliary loss contains three components. $\mathcal{L}_{\log}$ is a Huber loss between predicted and target log-depth over valid pixels. $\mathcal{L}_{\nabla}$ matches horizontal and vertical log-depth

TABLE I

SUCCESS RATE (%) ON LIBERO AND ROBOTWIN 2.0. EMB. PT. INDICATES ADDITIONAL EMBODIED/ROBOT-DATA PRETRAINING BEFORE TARGET-BENCHMARK TRAINING. BOLD AND UNDERLINED VALUES DENOTE THE HIGHEST AND SECOND-HIGHEST VALUES IN EACH BENCHMARK COLUMN, RESPECTIVELY.

Method	Emb. PT.	LIBERO					RoboTwin 2.0		
		Spatial	Object	Goal	Long	Avg.	Clean	Rand.	Avg.
$\pi_{0.5}$ [26], [6]	Yes	98.8	98.2	98.0	92.4	96.9	82.74	76.76	79.75
X-VLA [27], [3]	Yes	98.2	98.6	97.8	97.6	98.1	72.80	72.84	72.82
ABot-M0 [28]	Yes	98.8	<u>99.8</u>	99.0	96.6	<u>98.6</u>	80.42	81.16	80.79
StarVLA- α [29]	No	99.0	<u>99.8</u>	<u>98.5</u>	94.1	97.9	88.20	88.30	88.25
Motus [3], [6]	Yes	96.8	<u>99.8</u>	96.6	97.6	97.7	88.66	87.02	87.84
LingBot-VA [8]	Yes	98.5	<u>99.6</u>	97.2	98.5	98.5	92.93	91.55	92.24
FastWAM [6]	No	98.2	100.0	97.0	95.2	97.6	91.88	91.78	91.83
FastWAM-Joint [6]	No	<u>99.6</u>	99.4	98.2	96.8	98.5	90.84	90.32	90.58
LiLa-WAM [30]	No	98.0	98.8	97.2	94.2	97.1	90.48	89.04	89.76
MVG-WAM (Ours)	No	99.8	<u>99.8</u>	99.0	<u>97.8</u>	99.1	<u>92.54</u>	<u>91.60</u>	<u>92.07</u>

gradients at multiple spatial scales. $\mathcal{L}_{\text{bnd}}$ applies the gradient residual with an RGB-derived spatial weight.

The boundary weight is computed from the normalized grayscale gradient of the corresponding future ground-truth RGB frame. For video timestep τ_v , we use $w(\tau_v) = \max((1 - \tau_v/T)^\gamma, w_{\min})$ to reduce supervision at high noise levels.

The depth and overall objectives are

$$\begin{aligned}
 \mathcal{L}_{\text{aux}} &= \lambda_{\log} \mathcal{L}_{\log} + \lambda_{\nabla} \mathcal{L}_{\nabla} + \lambda_{\text{bnd}} \mathcal{L}_{\text{bnd}}, \\
 \mathcal{L}_{\text{depth}} &= \mathbb{E}[w(\tau_v) \mathcal{L}_{\text{aux}}], \\
 \mathcal{L} &= \lambda_{\text{vid}} \mathcal{L}_{\text{vid}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{dep}} \mathcal{L}_{\text{depth}}.
 \end{aligned} \tag{10}$$

The geometry encoder, video VAE, and text encoder remain frozen. The video and action experts are jointly optimized with the new MVG modules. We use separate learning rates for pretrained and newly initialized parameters.

During rollout, the geometry representation is computed once from the current synchronized observation and reused across denoising steps. The auxiliary depth decoder is disabled at inference.

IV. EXPERIMENTS

Our experiments address four questions:

Q1, how well does MVG-WAM perform in simulation and on a real robot?

Q2, how robust is it to distribution shifts?

Q3, which geometric components contribute to control?

Q4, how well does the learned representation support short-horizon prediction and metric multi-view geometry?

A. Experimental Setup

Data and evaluation. LIBERO [32] contains 40 tasks across Spatial, Object, Goal, and Long, with 50 demonstrations per task. RoboTwin 2.0 [33] uses 50 tasks with 50 clean and 500 randomized demonstrations per task. We additionally record synchronized simulator depth for auxiliary supervision. One policy is trained jointly across all LIBERO suites, and a separate policy covers all RoboTwin 2.0 tasks. Following FastWAM [6], we evaluate 50 trials per LIBERO task and 100 trials per RoboTwin 2.0 task and domain.

Training and inference. The video backbone is initialized from Wan2.2 [34]. The video and action experts are optimized end-to-end with the MVG modules, while the geometry encoder, video VAE, and text encoder remain frozen. Training uses eight H100 GPUs, bfloat16, a global batch size of 128, and AdamW [35]. Learning rates are 5×10^{-6} for pretrained parameters and 10^{-4} for newly introduced modules. LIBERO and RoboTwin 2.0 are trained for 30k and 100k updates, respectively. Inference uses ten denoising steps and 32-action chunks; the auxiliary depth decoder is not used during control.

B. Simulation Benchmark Results

To address Q1 in simulation, Table I compares MVG-WAM with representative VLA and WAM policies, while distinguishing whether additional embodied pretraining is used.

Without additional embodied pretraining, MVG-WAM achieves 99.1% average SR on LIBERO and 92.07% on RoboTwin 2.0. Compared with FastWAM-Joint, this corresponds to gains of 0.6 and 1.49 percentage points, respectively.

On RoboTwin 2.0, MVG-WAM also outperforms FastWAM by 0.24 points and remains only 0.17 points below the embodied-pretrained LingBot-VA. For reference, LingBot-VA reports 5.23 s under its RoboTwin 2.0 configuration [36], while our implementation takes 0.74 s per 32-step chunk on an H100. These results show that MVG-WAM attains competitive manipulation performance without additional embodied pretraining, while maintaining substantially lower inference latency than LingBot-VA.

The Clean-Randomized gap is 0.94 points. Since randomized demonstrations are included during training, this comparison does not constitute an OOD evaluation.

C. Out-of-Distribution Generalization

To address Q2, we evaluate two complementary distribution shifts: the unchanged LIBERO policy on LIBERO-

TABLE II

LIBERO-PLUS SUCCESS RATE (%) UNDER SEVEN DISTRIBUTION SHIFTS. OPENVLA AND OPENVLA-OFT RESULTS FOLLOW LIBERO-PLUS [37]; FASTWAM-JOINT [6] AND MVG-WAM USE OUR EVALUATION. BOLD AND UNDERLINED VALUES DENOTE THE HIGHEST AND SECOND-HIGHEST VALUES IN EACH ROW, RESPECTIVELY.

Shift	OpenVLA	OpenVLA-OFT	FastWAM-Joint	MVG-WAM
Camera	0.8	56.4	39.9	<u>51.9</u>
Robot	3.5	31.9	<u>65.1</u>	65.3
Language	23.0	79.5	94.7	<u>90.1</u>
Light	8.1	<u>88.7</u>	92.1	88.4
Background	34.8	93.3	58.1	<u>62.3</u>
Noise	15.2	75.8	56.2	<u>57.6</u>
Layout	28.5	74.2	<u>79.3</u>	80.8
Overall	15.6	<u>69.6</u>	68.7	70.4

TABLE III

ROBOTWIN 2.0 CLEAN2CLEAN (C2C), CLEAN2RANDOM (C2R), AND AVERAGE SUCCESS RATES (%). X-WAM, FASTWAM, AND AHA-WAM RESULTS FOLLOW THE ROBOTWIN 2.0 LEADERBOARD SNAPSHOT [40];

FASTWAM-JOINT AND MVG-WAM ARE EVALUATED IN OUR CLEAN-ONLY SETTING. BOLD AND UNDERLINED VALUES DENOTE THE HIGHEST AND SECOND-HIGHEST VALUES IN EACH ROW, RESPECTIVELY.

	X-WAM	FastWAM	AHA-WAM	FastWAM-Joint	MVG-WAM
C2C	70.0	77.8	64.3	69.8	<u>72.6</u>
C2R	<u>25.8</u>	1.9	3.2	1.3	29.2
Avg.	<u>47.9</u>	39.9	33.8	35.6	50.9

Plus [37], and clean-only RoboTwin 2.0 training followed by Clean2Random evaluation.

LIBERO-Plus. Table II compares our evaluation with published OpenVLA and OpenVLA-OFT results [38], [39]. MVG-WAM improves the pooled SR from 68.7% for FastWAM-Joint to 70.4%. The largest gains over FastWAM-Joint occur under Camera and Background shifts (+12.0 and +4.2 points), while performance decreases under Language and Light perturbations. The gains therefore do not extend uniformly across all shifts.

RoboTwin 2.0 Clean2Random. Following the official Clean2Random setting, we train FastWAM-Joint and MVG-WAM as multi-task policies on the same 2,500 clean demonstrations (50 per task) without exposure to randomized RoboTwin 2.0 data. Both policies are trained for 100k updates and evaluated over 100 trials per task in each test domain.

Table III additionally includes the X-WAM, FastWAM, and AHA-WAM results reported in the RoboTwin 2.0 leaderboard snapshot [40]. MVG-WAM reaches 72.6% C2C and 29.2% C2R SR, improving over FastWAM-Joint by 2.8 and 27.9 percentage points, respectively. The remaining gap between C2C and C2R indicates that unseen randomized conditions remain substantially more challenging.

D. Ablation Studies

To address Q3, Table IV isolates the geometric mechanisms of MVG-WAM on RoboTwin 2.0. For computationally

TABLE IV

ROBOTWIN 2.0 ABLATION SUCCESS RATES (%) UNDER THE REDUCED 50-CLEAN/200-RANDOM TRAINING REGIME. HARD-13 IS FIXED BY INDEPENDENT FASTWAM-JOINT VALIDATION. ROWS BEGINNING WITH “w/o” DENOTE MVG-WAM VARIANTS WITH THE CORRESPONDING COMPONENT REMOVED.

Variant	All 50	Hard 13
MVG-WAM (Ours)	92.05	77.65
w/o Epipolar Residual	91.05	75.00
w/o Epipolar Constraint	90.86	74.65
w/o View Routing	90.93	74.92
w/o Depth Supervision	90.96	74.81
FastWAM-Joint (Baseline)	89.44	71.96

efficient ablation, all variants are trained on the same 50 tasks using 50 clean and 200 randomized demonstrations per task and are evaluated with shared seeds. This reduced training regime is used only for the ablation study; the standard RoboTwin 2.0 results in Table I use 50 clean and 500 randomized demonstrations per task.

Hard-13 is fixed by an independent FastWAM-Joint validation pass using tasks with SR below 90%. The variant without the epipolar constraint retains the same dense encoder and retrieval budget but removes geometric restriction during cross-view retrieval, while the variant without view routing exposes every canvas region to the full per-view token bank. Because the ablation series uses a reduced training regime and is evaluated independently from the standard benchmark run, we report its corresponding full-model result rather than substituting the score from Table I.

The complete model improves over FastWAM-Joint by 2.61 points across all 50 tasks and 5.69 points on Hard-13. Removing the epipolar residual, epipolar constraint, view routing, or depth supervision consistently reduces SR. In particular, removing metric-depth supervision decreases performance by 1.09 points on all tasks and 2.84 points on Hard-13, supporting the benefit of metric-scale grounding for the MVG representation. Together, these controlled comparisons support the complementary roles of cross-view epipolar modeling, view-indexed routing, and metric-scale grounding.

E. Real-World Experiments

To address Q1 on a physical system, we evaluate MVG-WAM on an AgileX Cobot Magic using radish-to-plate, cup stacking, and pot opening followed by pepper placement. Figure 3 shows the platform and representative executions. The scene-view camera is wrist-mounted at a fixed observation angle.

We collect 100 RGB-D training and ten validation demonstrations per task and fine-tune one three-task policy for 20k updates. Each task is evaluated over 50 trials; interventions and safety stops count as failures.

Table V reports 96%, 90%, and 88% SR for MVG-WAM, yielding 137/150 successful trials (91.3%). Under the same evaluation protocol, $\pi_{0.5}$ achieves 92%, 88%, and 88% SR, corresponding to a mean SR of 89.3%.

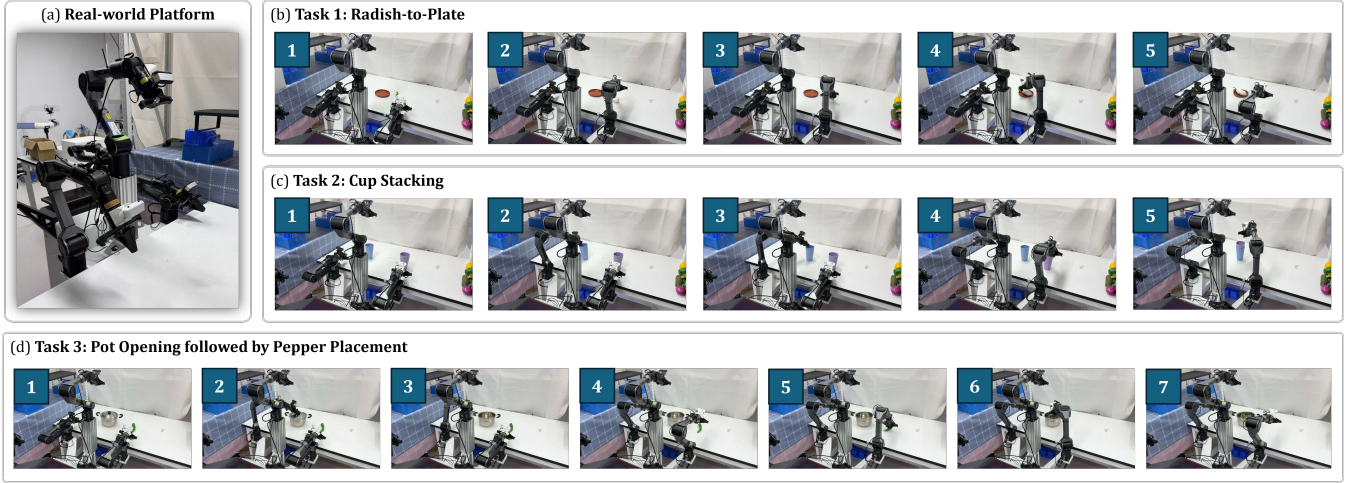


Fig. 3. **Real-world evaluation on Cobot Magic.** (a) Physical manipulation platform with the wrist-mounted scene-view camera. (b)–(d) Representative execution sequences for radish-to-plate, cup stacking, and pot opening followed by pepper placement, respectively. The third task consists of two stages: removing the pot lid and subsequently placing the pepper into the pot. Frames progress from left to right within each sequence. The shown frames are captured from an external recording viewpoint rather than from the camera observations supplied to the policy. Quantitative success rates over 50 trials per task are reported in Table V.

TABLE V

COBOT MAGIC SUCCESS RATE (%), 50 TRIALS PER TASK. R, C, AND O DENOTE RADISH-TO-PLATE, CUP STACKING, AND POT OPENING FOLLOWED BY PEPPER PLACEMENT. “W/O” DENOTES WITHOUT.

Model	R	C	O	Mean
$\pi_{0.5}$ [26]	92	88	88	89.3
FastWAM-Joint [6]	86	78	72	78.7
MVG-WAM w/o Epipolar Residual	94	86	82	87.3
MVG-WAM (Ours)	96	90	88	91.3

TABLE VI

SHORT-HORIZON RGB AND TEACHER-FORCED DEPTH DIAGNOSTICS ON ROBOTWIN 2.0, AVERAGED EQUALLY OVER CLEAN AND RANDOM DOMAINS. A DASH DENOTES NOT APPLICABLE; FASTWAM-JOINT HAS NO TRAINED NATIVE DEPTH HEAD. STARRED GT ENTRIES ARE ANALYTICAL SELF-COMPARISONS; CV AND COVERAGE ARE REPROJECTION REFERENCES.

Model / ref.	MAE↓ (cm)	AbsRel↓ (%)	CV↓ (cm)	Cov. (%)	PSNR↑ (dB)	SSIM↑
FastWAM-Joint	—	—	—	—	28.05	0.842
MVG-WAM (Ours)	2.12	5.80	1.68	16.70	31.79	0.914
GT reference	0.00*	0.00*	0.25	19.00	∞^*	1.000*

The real-world results further support Q3: removing the epipolar residual reduces mean SR by 4.0 points.

The controller executes 24-waypoint chunks at 20 Hz; a batch-size-one policy call takes 0.81 s on an H100, including preprocessing and geometry encoding.

F. Multi-View Geometric Analysis

To address Q4, we evaluate short-horizon RGB prediction and teacher-forced (TF) metric-depth readout on 625 clips from 125 trajectories in the existing RoboTwin 2.0 validation split.

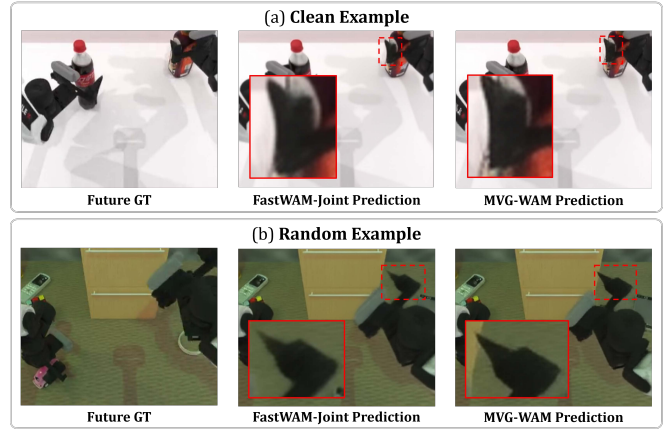


Fig. 4. Qualitative short-horizon RGB prediction on RoboTwin 2.0. FastWAM-Joint and MVG-WAM use the same current observation and are compared against the same future target. Dashed boxes indicate gripper-object interaction regions, with enlarged crops shown in red. MVG-WAM produces more faithful local predictions, preserving sharper object boundaries and more coherent gripper-object geometry.

RGB prediction uses only the current observation, proprioception, and instruction context. The depth readout is evaluated separately from noisy future latents and therefore serves as a diagnostic of the learned representation rather than current-observation-only depth forecasting. For compactness, Table VI reports the equal average of the previously computed Clean and Random task-macro scores. FastWAM-Joint has no trained native depth head, so its corresponding depth metrics are not applicable.

MVG-WAM improves average RGB PSNR by 3.74 dB and SSIM by 0.072 over FastWAM-Joint. Figure 4 provides a qualitative comparison of the corresponding short-horizon predictions. The improvement is particularly visible around gripper-object interaction regions, where MVG-WAM better

preserves local object boundaries and gripper-object structure. The enlarged crops highlight these manipulation-critical regions while both methods are compared against the same future target.

For the TF depth diagnostic, we corrupt future target latents at a fixed video diffusion timestep $\tau_v = 400$ before reading out metric depth. Under this setting, the averaged TF depth readout of MVG-WAM reaches 2.12 cm MAE, 5.80% AbsRel, and 1.68 cm cross-view z error. Its reprojection coverage is 16.70%, compared with 19.00% for the GT reference; coverage is therefore interpreted together with reprojection error rather than as a stand-alone accuracy measure. These depth diagnostics indicate that the denoising representation retains metric geometric information across views and future horizons. Since FastWAM-Joint has no trained depth head, they are not interpreted as a direct depth-prediction improvement over the baseline.

V. CONCLUSION

We presented MVG-WAM, a video-based world-action model that treats synchronized camera observations as geometrically related projections of one physical world. By combining an epipolar-constrained global state with camera-aware routing, the model integrates cross-view evidence while retaining view-indexed geometric context. Future metric-depth supervision metrically grounds this geometry-aware representation as the scene evolves, without requiring depth decoding during action rollout.

Without additional embodied pretraining, MVG-WAM achieves average success rates of 99.1% on LIBERO and 92.07% on RoboTwin 2.0. Real-world evaluation further achieves 91.3% success across 150 trials on Cobot Magic. Together, these results demonstrate that explicitly organizing multi-view geometry provides an effective representation for world-action modeling in robotic manipulation.

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